

2	(a) (i) $27.2 + 273.15$ or $27.2 + 273.2$ 300.4 K	C1 A1 [2]
	(ii) 11.6 K	A1 [1]
	(b) (i) $\langle c^2 \rangle$ is the mean / average square speed	B1 [1]
	(ii) $\rho = Nm/V$ with <u>N explained</u> so, $pV = 1/3 Nm\langle c^2 \rangle$ and $pV = NkT$ with <u>k explained</u> <u>so mean kinetic energy / $\langle E_K \rangle = 1/2 m\langle c^2 \rangle = 3/2 kT$</u>	B1 B1 B1 B1 [4]
	(c) (i) $pV = nRT$ $2.1 \times 10^7 \times 7.8 \times 10^{-3} = n \times 8.3 \times 290$ $n = 68 \text{ mol}$	C1 A1 [2]
	(ii) mean kinetic energy = $3/2 kT$ = $3/2 \times 1.38 \times 10^{-23} \times 290$ = $6.0 \times 10^{-21} \text{ J}$	C1 A1 [2]
	(iii) realisation that total internal energy is the total kinetic energy energy = $6.0 \times 10^{-21} \times 68 \times 6.02 \times 10^{23}$ = $2.46 \times 10^5 \text{ J}$	C1 C1 A1 [3]

2	(c) (i) to and fro / backward and forward motion (between two limits)	B1 [1]
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